RIN RAY

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CRC Deformations-Geometry-Structure Postdoctoral Fellow, Universität Münster

Eisenstein series · zeta functions and special values · the Lubin–Tate action

ACADEMIC EMPLOYMENT

2023–Present — **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger

- chromatic homotopy theory · Eisenstein series and categorical traces · genera and equivariant K-theory · zeta functions and special values · algebraic K-theory

EDUCATION

2017–2023 — **Northwestern**; PhD in Maths; advised by Paul Goerss

- *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*

2015–2017 — **UChicago**; Divisional Masters Program in Maths; advised by Peter May and Kazuya Kato

- *Calculating $\pi_*(tmf)$ at the prime 2*
- *Models of Formal Group Laws of Every Height*

2012–2013 — **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos

- *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
- *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*

2010–2012 — **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

RESEARCH

Sept 2026 — **On the Chromatic Splitting Conjecture in Coheight 1**; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; coheight-one weak chromatic splitting at every height, for generic primes; before, known only at heights 1 and 2

Sept 2025 — **Tiny and Unique: The Lubin–Tate Action at Height $p-1$ from the Minimal Curve with Its Symmetries** submitted: *AG&T*; arXiv:2509.23428

V3 OCT 2026

July 2025 — **Towards the $p = 3$ Kervaire Invariant Problem** submitted; joint with E. Belmont; the E_2-page for the homotopy fixed points spectral sequence computing $\pi_*(E_6^{hC_9})$; arXiv:2507.10157 · both our results and the method we proposed for the remaining case were used in a claimed resolution of the $p = 3$ Kervaire invariant problem (OpenAI preprint, Sept 2026)

July 2025 — **Modeling Group Actions on Stacks (Especially the Lubin–Tate Action)** submitted: *Journal of Geometry and Physics*; arXiv:2507.00309

V2 OCT 2026

2022 — **Automorphisms of Abelian Varieties and Principal Polarizations** published; joint with D. Lee; *Rendiconti del Circolo Matematico di Palermo*, Series 2, vol. 71, pp. 483–494; arXiv:1811.07007

2020 — **Towards Directed Collapsibility** published; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; *Advances in Mathematical Sciences*, vol. 21, pp. 255–271; arXiv:1902.01039

2018 — **A Global Crystalline Period Map**; joint with M. Neaton and A. Pieper; from the Arizona Winter School; arXiv:1911.08615

CURRENT PROJECTS

Symmetry — **Only the Gauge Grows: Moduli Stacks of G-Curves in Homotopy Theory at $h = p^{k-1}(p-1)$**; Part III

Scattering — **The Unramified Eisenstein Spectrum as a Categorical Trace**; solo

Echoes	**The λ-twisted $T(h+1)$-local Zeta Function**; joint with G. Angelini-Knoll
Echoes	**Toward Categorifying Symplectic L-functions and Reidemeister Torsion**; joint with A. Petrov and M. Volpe
Echoes	***L*-functions and the Relative Image of *J* in Moore Spectra**
Echoes	**All Bernoulli Numbers in Homotopy Theory Are Shifts**; joint with A. Mejia and N. Riggenbach

FELLOWSHIPS AND AWARDS

2022	**Gelfand Prize**, Northwestern University; for outstanding progress in one's own research and extraordinary contributions to the research goals of the graduate community
2017	Awarded **NSF GRFP Fellowship** (103K + 36K)
2014–2016	Awarded **Thiel Fellowship** (100K); mathematics and assistive technology; mentor Edward Frenkel

TEACHING

2025	**Mathe trifft Kunst** (Math Meets Art), Münster; designed hands-on math workshops for children
2017–2023	**Teaching Assistant**, Northwestern; discussion sections, quizzes and solutions: Intensive Linear Algebra and Multivariable Calculus (MATH 291, honors), Linear Algebra (MATH 240), Discrete Mathematics · evaluations (2021): 5.65/6 interest in teaching, 5.48/6 clarity
2020–2023	**Directed Reading Program**, Northwestern; weekly one-on-one reading with an undergraduate, ending in a talk to their peers
2015–2017	**Research Experience for Undergraduates**, University of Chicago; weekly lectures, and guiding each student through an individual research project
2016–2017	**Math Circles**, South Side of Chicago; weekly mathematics with children aged 6–12
2012	**Teaching Assistant, College Algebra**, Mary Baldwin College

COMMUNITY AND OUTREACH

2024	**Panelist, Queer in Math Day**, MPI Leipzig
2021–2023	**Causeway Postbaccalaureate Program** and **TGS Diversity Peer Mentorship**, Northwestern; mentoring students from diverse backgrounds into and through graduate school
2021–	**Founder, Mental Health Math Buddy Program**, Northwestern; pairs first- and second-year graduate students with older students for quarterly check-ins; ongoing
2022	**Established the Fund for Work–Life Activities for Graduate Students**, Northwestern
2020–2023	**Evanston Animal Shelter**; fostered special-needs adult cats and raised neonatal kittens
2014–	**Founding contributor, Topos House**, South San Francisco; a house for collaboration between mathematicians, biologists and artists; ongoing

SEMINARS ORGANIZED

2023–2025	**Universität Münster**; Homotopy and Arithmetic working groups: K-Theory of the Integers · Euler Characteristics and Zeta Functions · K-Theoretic Poitou–Tate Duality · Linearity in p-adic Lie Theory; and the Young Topologists Meeting (2024)
2018–2022	**Northwestern University**; Factorization Homology · Geometry for Prime Addicts (toward Scholze's proof of the weight-monodromy conjecture) · Formal Moduli Problems · TamagaWHAT (the Tamagawa conjecture for function fields) · Applications of Topos Theory · Moduli Spaces · Stop Staring and Compute!

SELECTED INVITED TALKS

2026	**Mod *p* Langlands and the Chromatic Splitting Conjecture**; Essen Number Theory Seminar
2024	**Moduli Stacks of Curves in Homotopy Theory**; Johns Hopkins Topology Seminar

2021 **Galois Theory of Spheres**; Front Range Number Theory

2020 **Explicit Arithmetic of Jacobians**; ICERM Workshop on Arithmetic Geometry, Number Theory, and Computation

2019 **Inverse Galois Theory in the Homotopy Groups of Spheres**; Princeton CCR Colloquium

2019 **Galois Theory and Galois Representations in Homotopy Theory**; Higher School of Economics, Moscow

2019 **An Introduction to the Classic Theory of *p*-Divisible Groups**; Oberwolfach Arbeitsgemeinschaft

RESEARCH OUTSIDE MATHEMATICS

2026 **Memantine as a candidate non-opioid analgesic in central sensitization**; NMDA receptor antagonism · opioid tolerance and hyperalgesia · drug repurposing · trial design

2026 **Neostigmine in feline idiopathic megacolon**; enteric motility and cholinergic transmission · documentation of an unpublished treatment

2024– **Biome of Overstimulation: Sensory Sensitivity as a Unifying Mechanism**; joint with L. Estinto; sensory processing as an organizing principle · gut microbiome · autoimmunity · chronic pain · glutamate excitability

2015 **Dependent and directed type systems for multiphysics models**; Santa Fe Institute; multiscale modeling · compositional model building · topology applied to complex biological systems

2014 **Decoder convergence and optical recording for intracortical motor prostheses**; intracortical brain–computer interfaces · foreign body response and chronic signal loss

2013–2015 **Modular robotic retrofits for powered wheelchairs**; Thiel Fellowship; automated pressure redistribution · assisted transfer · gaze-controlled assistive mobility · user-led requirements

2013–2014 **Unsupervised classification of rodent ultrasonic vocalization for continuous preclinical monitoring**; Vium; computational bioacoustics · preclinical safety pharmacology · affective-state readout

2013 **Unsupervised grammar and lexicon induction from partially parallel text**; George Mason University; decipherment of incompletely understood scripts · contracted Braille as the test language

2013 **Automated abstractive summarization of biomedical literature for clinical readers**; George Mason University; computational semantics · evidence fidelity

2012–2013 **Electronic-structure prediction of superconducting transition temperature**; George Mason University, with D. Papaconstantopoulos; augmented plane wave method · intermetallic compounds · computational materials screening

2012 **Graceful degradation under sensor failure: contact-rich manipulation on proprioception alone**; George Washington University; reinforcement learning · Willow Garage PR2 · human–robot interaction

2012 **Monocular visual SLAM with motion planning on a micro aerial vehicle**; George Mason University; loop closure · scale from sonar and inertial fusion · motion planning on a partial map

2012 **Standing-wave mode structure of an RF discharge in a polyhedral cavity**; 2.45 GHz argon discharge · deployable structures in ionospheric plasma

2012 **Flexible TiO_2 resistive-switching devices**; Mary Baldwin College; memristive device fabrication · bend-cycle characterization · in-memory computing

2011–2014 **Portable detection of gluten and common allergenic proteins**; laser absorption spectroscopy · antibody colorimetric assay · immunoassay cost and sampling limits

CREATIVE WORKS

2025–2026 ***Impaction***, first play; staged in Los Angeles as part of *Nothing to See Here*, directed by Jacques Manjarrez · ***Endomortis***, comic; with Petra Flurin; crayolapunk surrealist comic about grief